

Question 1 (20 Marks) Select the best correct answer and write it in the table below.

1. The fuse symbol shown in the figure below is for: 
 - A. IEEE/ANSI standard
 - B. IEC standard
 - C. IEEE/ASNI standard
 - D. Old symbol
2. During the normal operation of the circuit, the fuse wire is:
 - A. A very low resistance component
 - B. A very high resistance component
 - C. A very high inductive component
 - D. A very low inductive component
3. The amount of current that a fuse can easily conduct without interrupting the circuit is known as:
 - A. Breaking capacity
 - B. I^2T value
 - C. Response characteristics
 - D. Current carrying capacity
4. The electronic fuse is a type of:
 - A. Surface Mount Technology (SMT) fuse
 - B. Bolted type fuse
 - C. High Rupturing Capacity (HRC) fuse
 - D. Rewirable type fuse
5. When a fault occurs in the system, the type of fuse that automatically resets after operation is classified as:
 - A. Current limiting fuse
 - B. Current limiting fuse
 - C. Reusable fuse
 - D. One-time-only fuse
6. The type of fuse used to protect transformers as an expulsion type fuse is:
 - A. Reusable fuse type
 - B. Dropout fuse type
 - C. Fast-acting chip fuse type
 - D. High-current-rated chip fuse type
7. For an electrical load of **1760 W** connected to a **220 V** supply, the required fuse rating will be:
 - A. 10 A
 - B. 8 A
 - C. 12 A
 - D. 17.6 A
8. The smallest current that will cause the fuse element to melt is:
 - A. Maximum using current
 - B. Minimum fusing current
 - C. Current rating
 - D. Fusing factor
9. For circuit breakers, the length of the arc is directly proportional to:
 - A. Charge capacitor
 - B. Magnetic field
 - C. Voltage
 - D. Cross-section
10. For air circuit breakers under fault current:
 - A. The main contacts open and the arcing contacts open.
 - B. The main contacts close and the arcing contacts open.
 - C. The main contacts open and the arcing contacts close.
 - D. The main contacts close and the arcing contacts close.

Answer Key

Question 1 2 3 4 5 6 7 8 9 10

Answer B A D A C B A B C C

Question 2 (5 Marks)

Mention the classification of LV fuses depending on the fusing factor.

Answer

Fuse Type Fusing Factor (FF)

FF Type $(1.75 \leq FF \leq 2.5)$

Q2 Type $(1.50 \leq FF \leq 1.75)$

Q1 Type $(1.25 \leq FF \leq 1.50)$

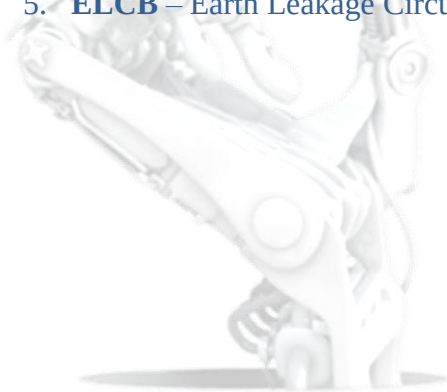
P Type $(FF \leq 1.25)$

Question 3 (5 Marks)

Mention five types of Low-Voltage AC circuit breakers.

Answer

1. **MCCB** – Molded Case Circuit Breaker
2. **MCB** – Miniature Circuit Breaker
3. **RCCB** – Residual Current Circuit Breaker
4. **GFCI** – Ground Fault Circuit Interrupter
5. **ELCB** – Earth Leakage Circuit Breaker



M E C H A T R O N I C S
S P E C T R U M

T E A M